



Virtual Mineral Kit



Overview

- Minerals are:
 - Naturally occurring
 - Inorganic (generally)
 - Solid
 - Have an orderly crystalline structure
 - Have a definite chemical composition (that allows for some variation)



Overview

- Minerals are identified by observing and testing physical properties:
 - Luster
 - Color – overall color of mineral
 - Hardness
 - Streak color – color of mineral when powdered (if softer than ceramic streak plate)
 - Crystal habit or cleavage



Mineral Identification: Luster

- Metallic – mineral reflects light (shiny like metal)
- Non-metallic – mineral does not look shiny like metal; instead, it is:
 - Glassy, waxy, pearly, earthy (dull), or greasy



Examples of metallic minerals: galena and pyrite

Mineral images from www.geologyscience.com



Examples of non-metallic minerals: quartz and orthoclase

Mineral images from www.geology.com



Mineral Identification: Hardness

	Mohs Scale of Hardness	Mineral Examples	Hardness of Some Common Objects (Harder objects scratch softer objects)
Hard	10	Diamond	6.5 Ceramic streak plate 5.5 Glass
	9	Corundum	
	8	Topaz	
	7	Quartz	
	6	Orthoclase	
Soft	5	Apatite	2.5 Fingernail
	4	Fluorite	
	3	Calcite	
	2	Gypsum	
	1	Talc	

Example #1:

A mineral sample scratches glass and leaves a powder on a streak plate. What is the hardness value of this mineral?

This mineral would be harder than glass (>5.5) and softer than the streak plate (<6.5).

Therefore, its hardness = 6

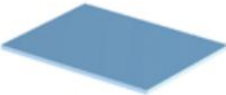
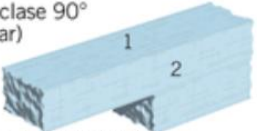
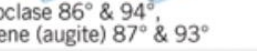
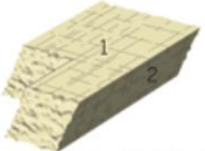
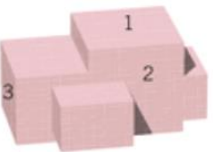
Example #2:

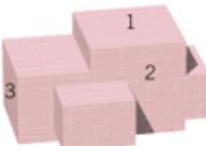
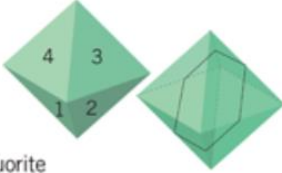
A mineral sample does not scratch glass and is not scratched by a fingernail. What is the hardness value of this mineral?

This mineral would be softer than glass (<5.5) and harder than a fingernail (>2.5). Therefore, its hardness ranges from 3 - 5



Mineral Identification: Cleavage / Fracture

Number of Cleavages and Their Directions	Name and Description of How the Mineral Breaks	Shape of Broken Pieces (cleavage directions are numbered)	Illustration of Cleavage Directions
No cleavage (fractures only)	No parallel broken surfaces; may have conchoidal fracture (like glass)	 Quartz	None (no cleavage)
1 cleavage	Basal (book) cleavage "Books" that split apart along flat sheets	 Muscovite, biotite, chlorite (micas)	
2 cleavages intersect at or near 90°	Prismatic cleavage Elongated forms that fracture along short <i>rectangular</i> cross-sections	 Orthoclase 90° (K-spar)  Plagioclase 86° & 94°, pyroxene (augite) 87° & 93°	
2 cleavages do not intersect at 90°	Prismatic cleavage Elongated forms that fracture along short <i>parallelogram</i> cross-sections	 Amphibole (hornblende) 56° & 124°	
3 cleavages intersect at 90°	Cubic cleavage Shapes made of cubes and parts of cubes	 Halite, galena	

3 cleavages intersect at 90°	Cubic cleavage Shapes made of cubes and parts of cubes	 Halite, galena	
3 cleavages do not intersect at 90°	Rhombohedral cleavage Shapes made of rhombohedrons and parts of rhombohedrons	 Calcite and dolomite 75° & 105°	
4 main cleavages intersect at 71° and 109° to form octahedrons, which split along hexagon-shaped surfaces; may have secondary cleavages at 60° and 120°	Octahedral cleavage Shapes made of octahedrons and parts of octahedrons	 Fluorite	
6 cleavages intersect at 60° and 120°	Dodecahedral cleavage Shapes made of dodecahedrons and parts of dodecahedrons	 Sphalerite	



Objectives

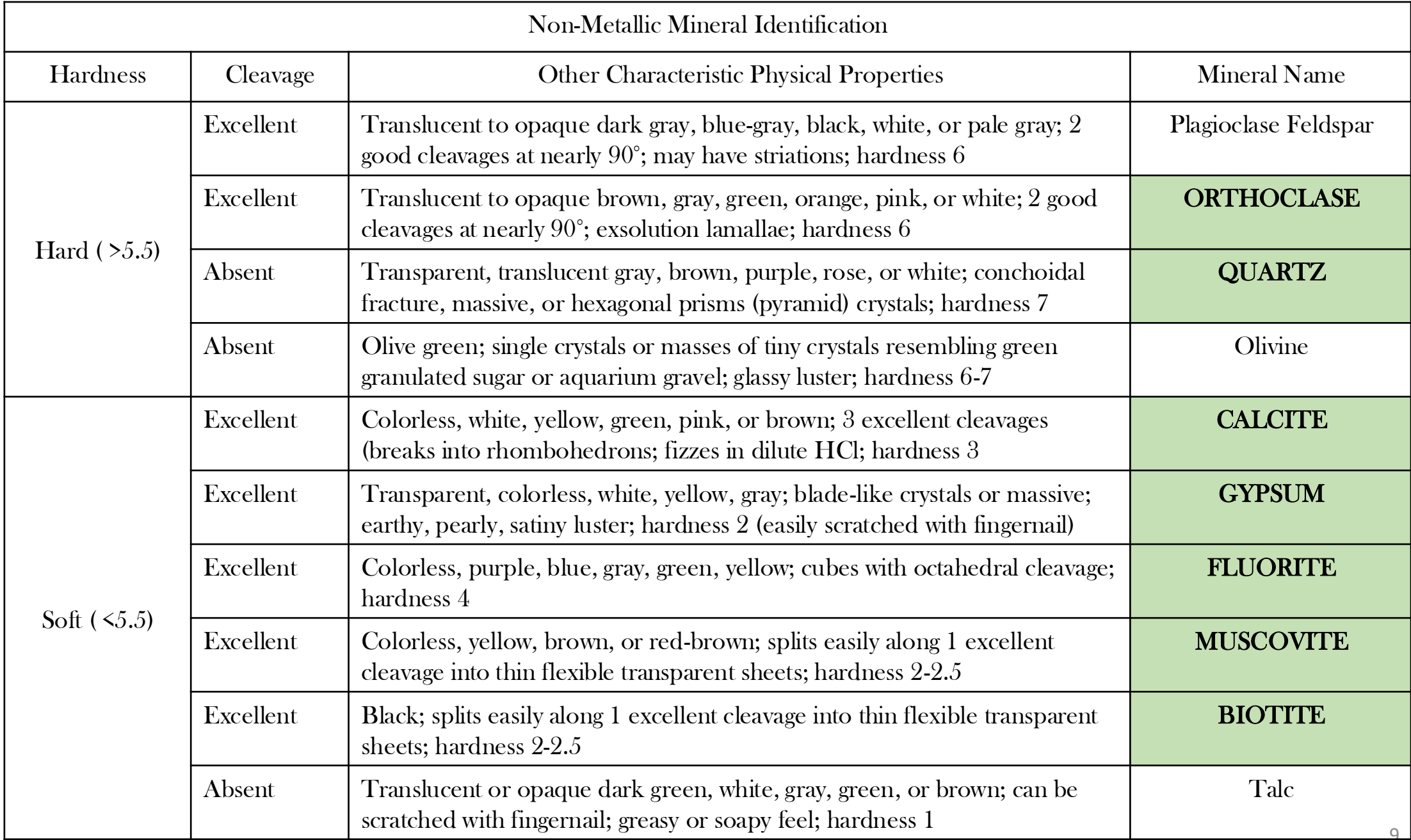
- Characterize the physical properties of the minerals in your virtual mineral kit
- Identify their proper name using the provided identification flow chart
 - The minerals highlighted and in bold uppercase font on the provided identification flow chart are provided in this virtual mineral kit:

EXAMPLE



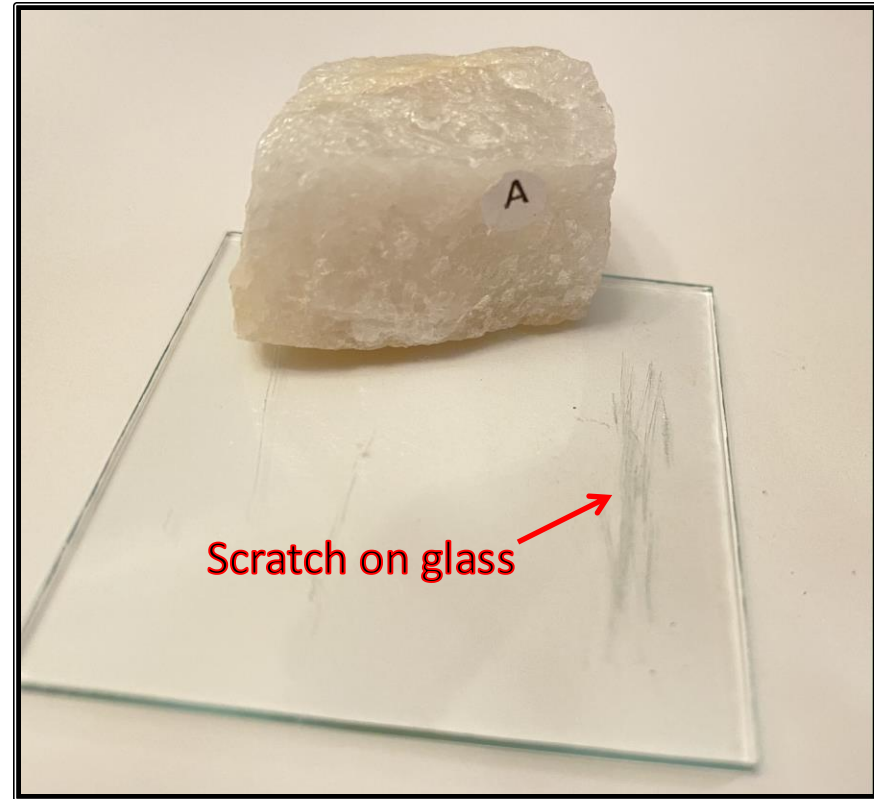
Mineral Identification Flow Charts

Metallic Mineral Identification				
Hardness	Cleavage	Streak Color	Other Characteristic Physical Properties	Mineral Name
Hard (>5.5)	Poor to good	Dark gray to black	Color silvery gold; hardness 6-6.5	PYRITE
	Absent	Dark gray to black	Color silvery dark gray to black; strongly attracted to a magnet; hardness 6-6.5	Magnetite
Soft (<5.5)	Good to excellent	Dark gray to black	Color bright silvery gray; breaks into cubes; hardness 2.5; feels heavy (high density)	GALENA
	Good to excellent	White to pale yellow-brown	Color silvery yellow-brown, silvery red, or black; hardness 3.5-4.0; smells like rotten eggs when powdered	Sphalerite
	Absent	Dark gray to black	Color dark silvery gray to black; can be scratched with fingernail - easily rubs off on finger; hardness 1-2	Graphite





Mineral Sample “A”



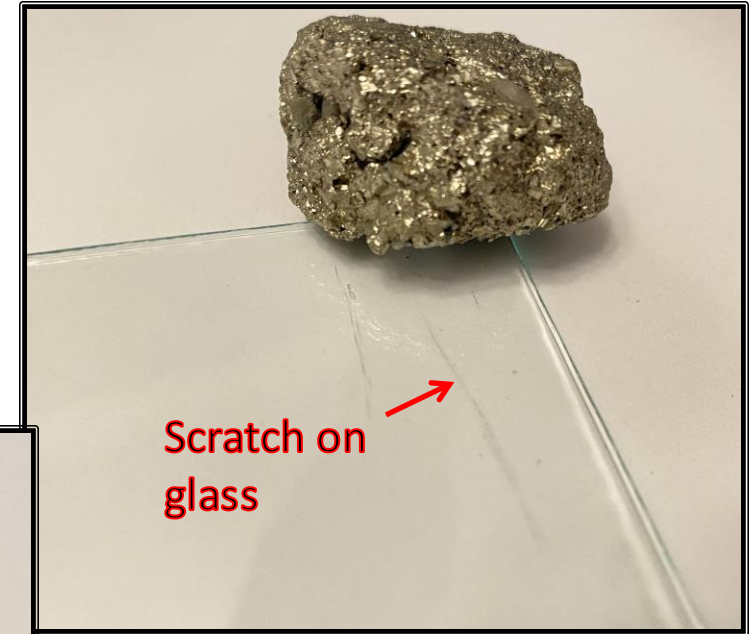
Cleavage

Yes

No



Mineral Sample “B”



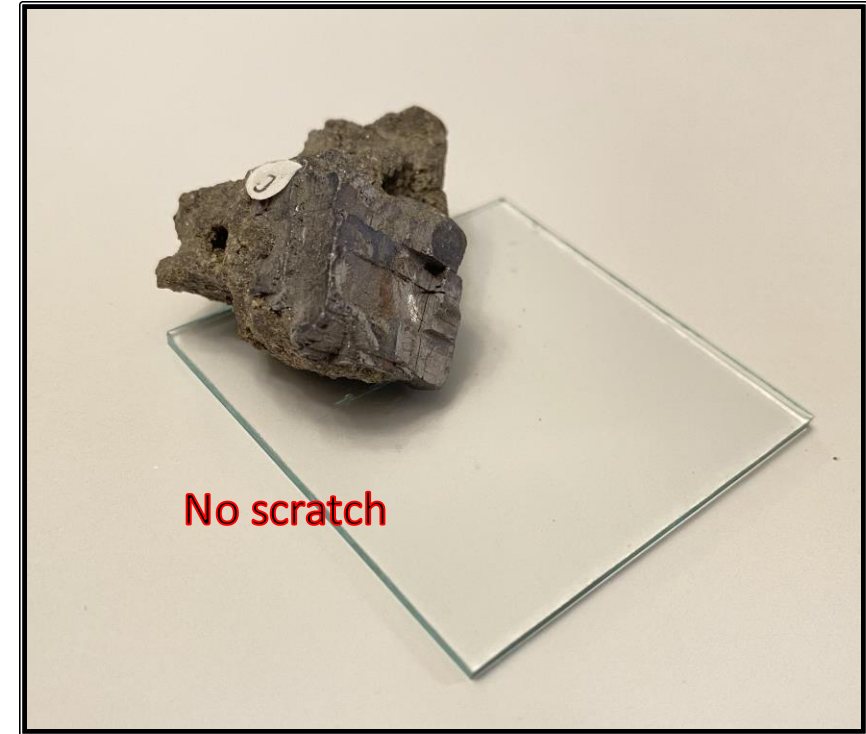
Cleavage

Yes

No



Mineral Sample “C”



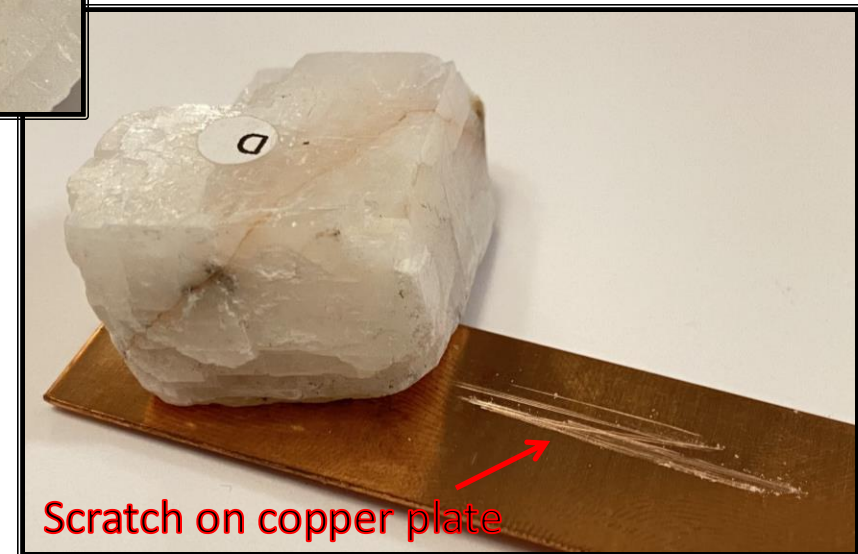
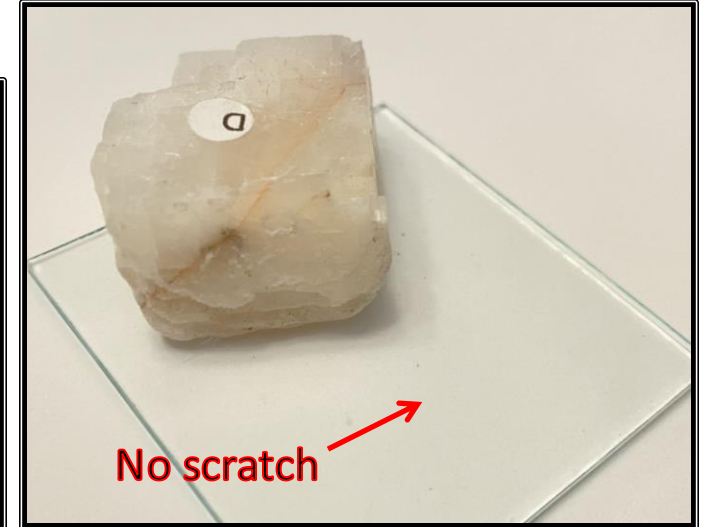
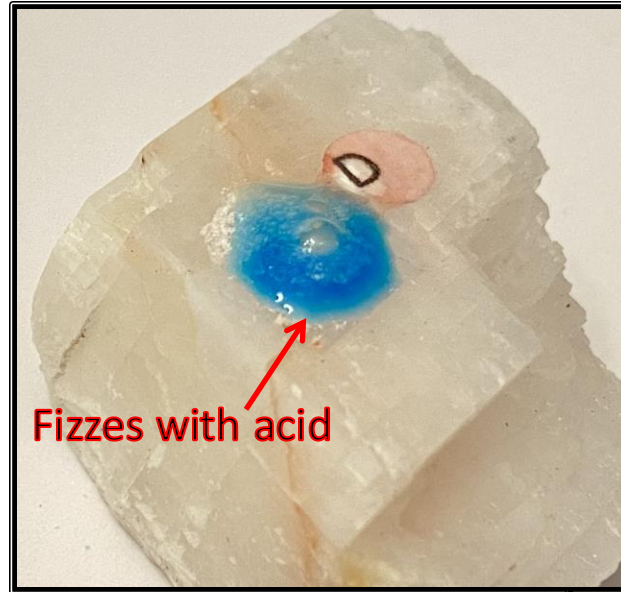
Cleavage

Yes

No



Mineral Sample “D”



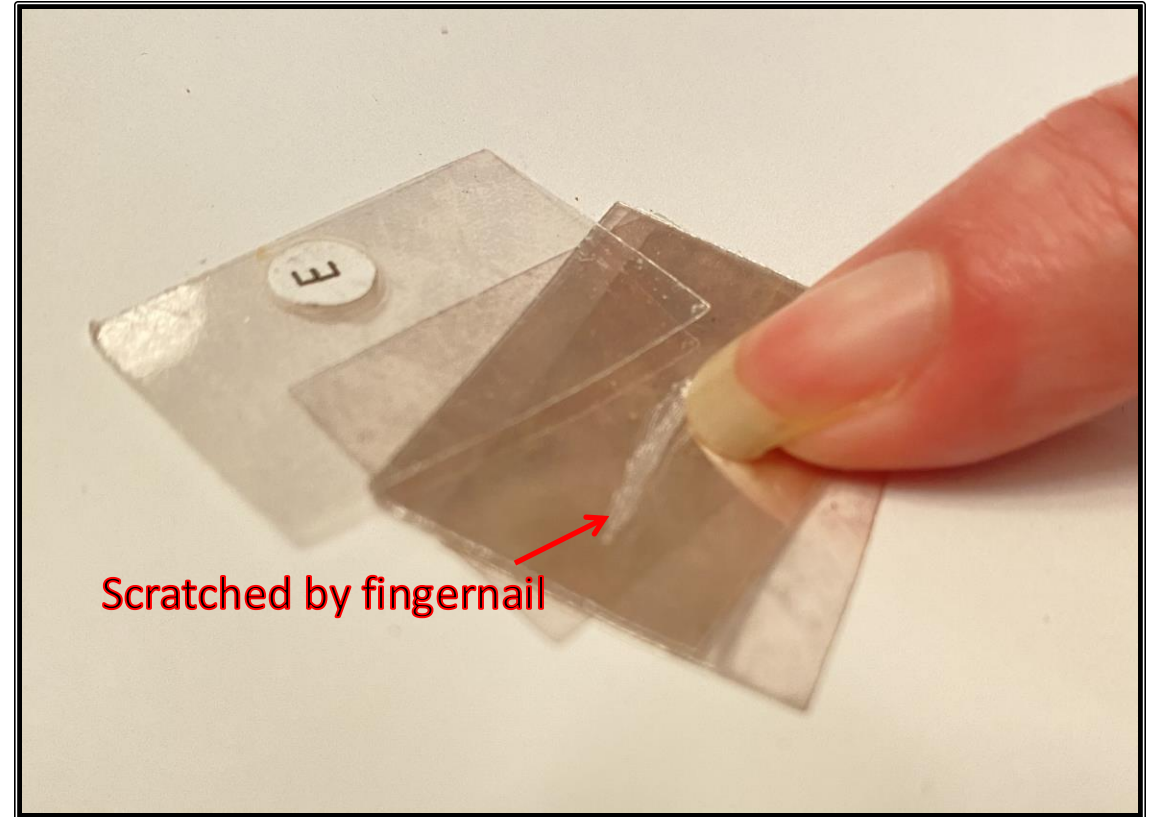
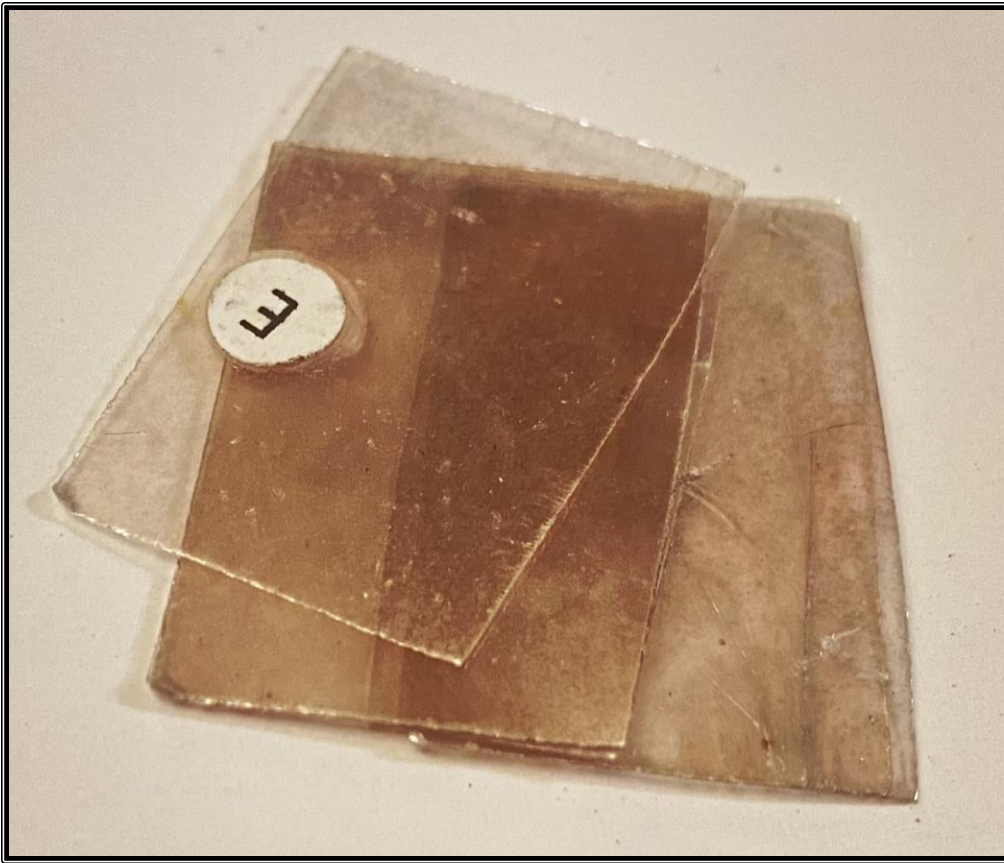
Cleavage

Yes

No



Mineral Sample “E”



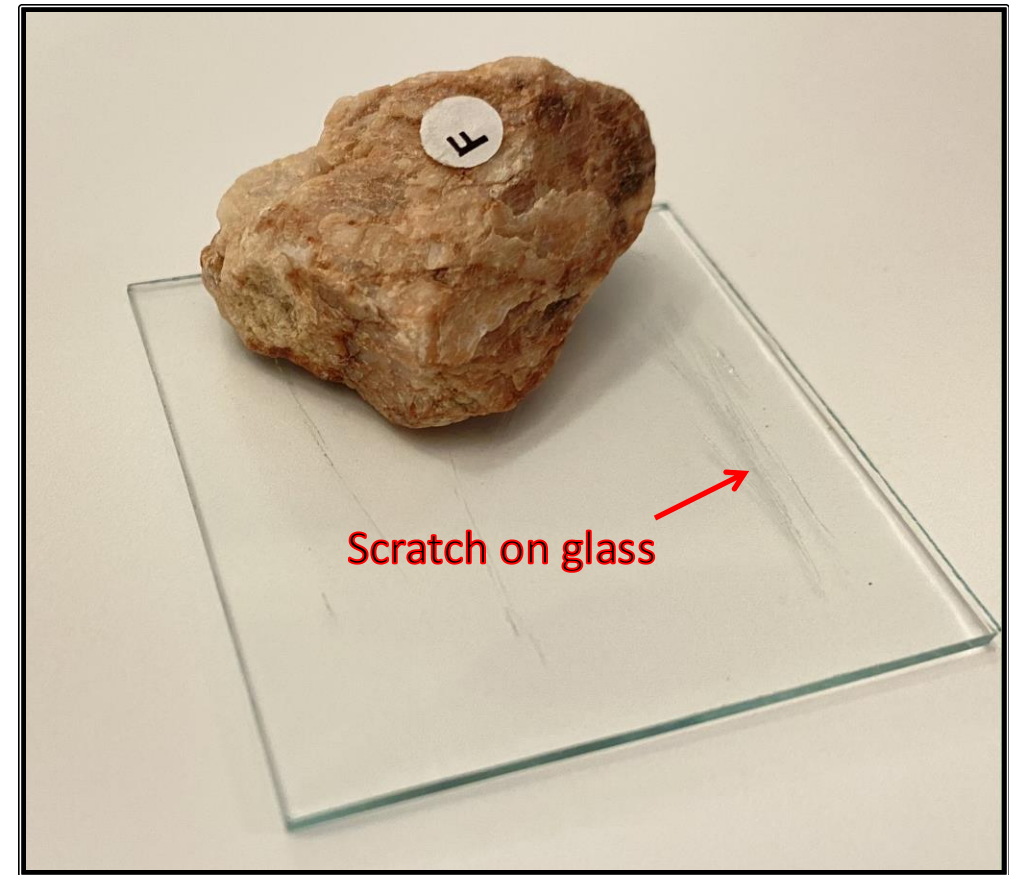
Cleavage

Yes

No



Mineral Sample “F”



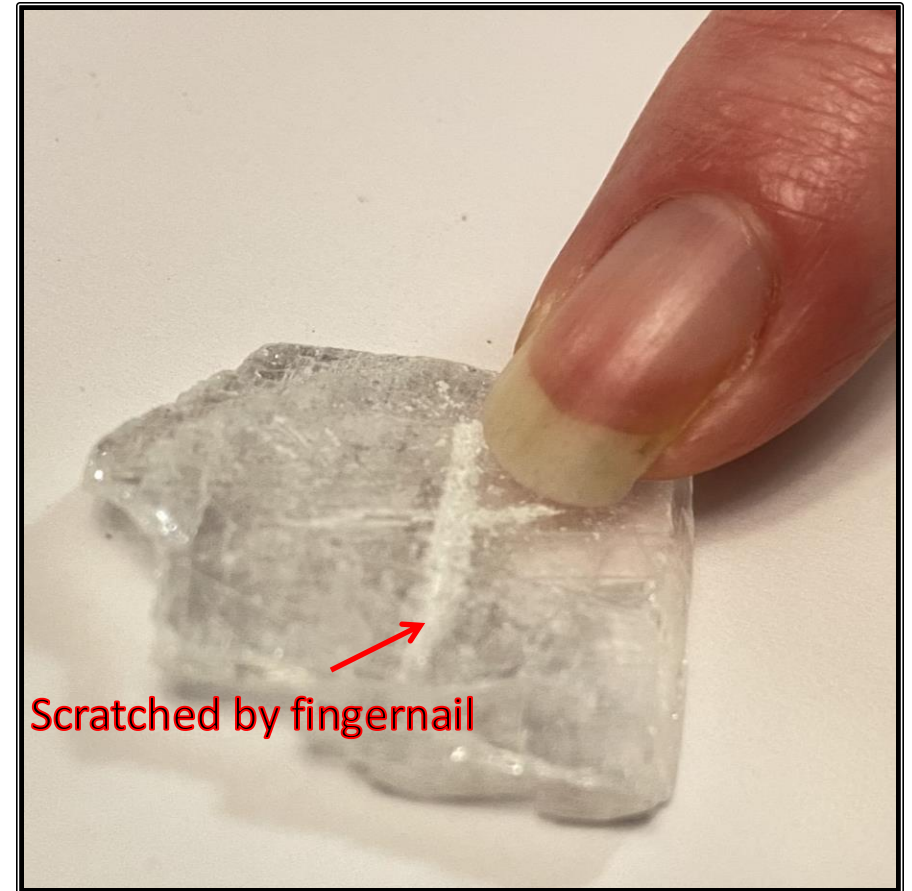
Cleavage

Yes

No



Mineral Sample “G”



Cleavage

Yes

No



Mineral Sample “H”



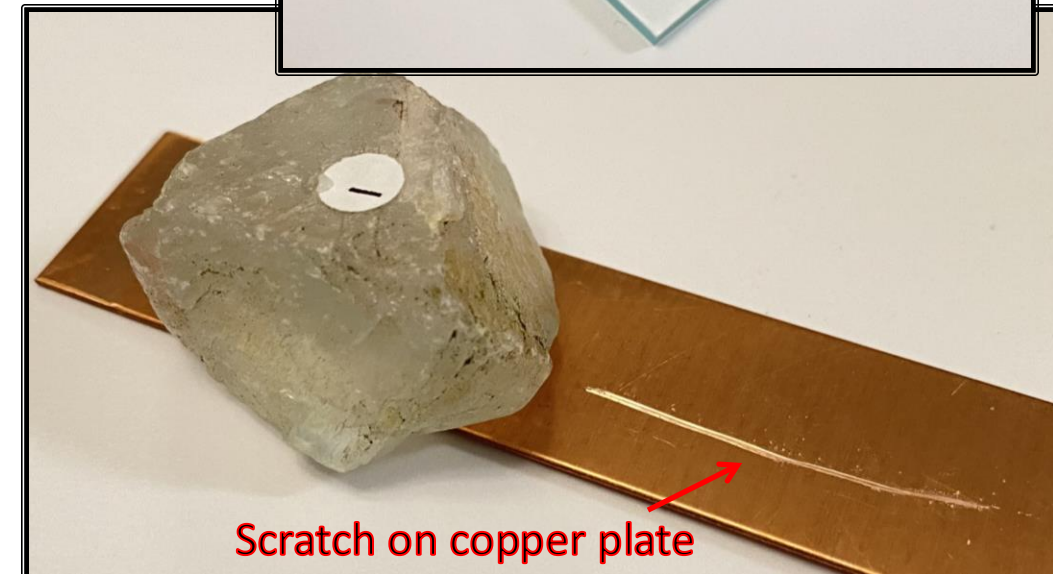
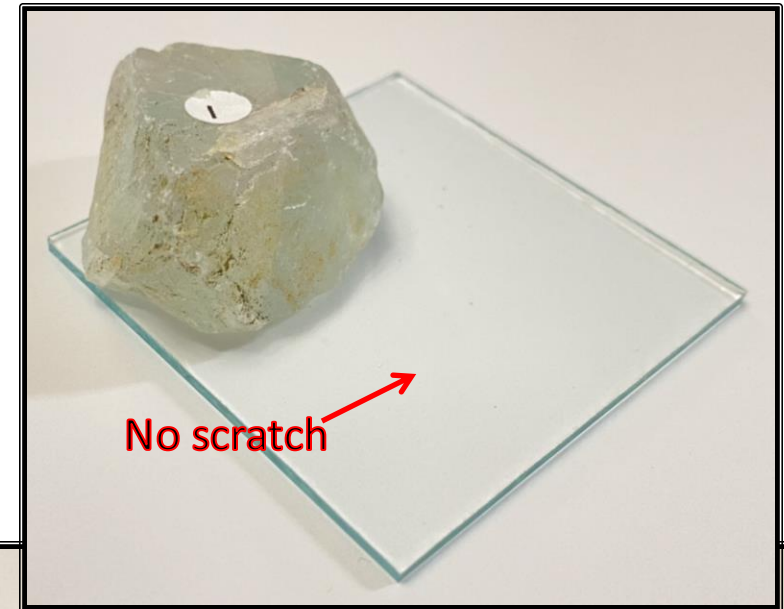
Cleavage

Yes

No



Mineral Sample “I”



Cleavage

Yes

No